

### Homework problems:

**Exercise 7.1** (8 points). Answer all the questions from the Quiz **Probability**.

*The quiz is available from May 25, 19:00 until **June 01, 09:00** . You have 60 minutes to answer the questions.*

**Remark:** The points for this problem count for the grade bonus.

**Solution:** To see the solution to this problem, please reopen the quiz and click on *Review*.

**Exercise 7.2.** The random variable  $X$  models the quality index of a product, with high values of  $X$  indicating high quality. Further let

$$f(x) = cx(2 - x)\mathbf{1}_{[0,2]}(x)$$

be a function, which depends on a parameter  $c$ .

- a) Show that  $f$  is the density of a distribution just for  $c = \frac{3}{4}$ .
- b) Show that

$$F(x) = x^2 \left( \frac{3}{4} - \frac{x}{4} \right) \mathbf{1}_{[0,2]}(x) + \mathbf{1}_{(2,\infty)}(x)$$

is the distribution function to the density  $f$ .

- c) Assume that  $f$  is the density of  $X$ . Compute the expectation of  $X$ .
- d) Products with a quality index of less than 0.15 are considered to be rejects. The manufacturer copes with a reject rate of 1% of his production. Is the manufacturer forced to sell products that are officially considered rejects? Justify your answer.

*Hint:* You can use the function `polyroot()` to come up with a justification in part d).

**Solution:**

- a) We need to show that  $f$  is a density function for  $c = \frac{3}{4}$ . Remember that  $f$  is a density if the conditions
  - i)  $f(x) \geq 0$  for all  $x \in \mathbb{R}$
  - ii)  $\int_{\mathbb{R}} f(x)dx = 1$are fulfilled.

Since

$$f(x) = \frac{3}{4}x(2 - x)\mathbf{1}_{[0,2]}(x) \geq 0 \quad \forall x \in \mathbb{R}$$

and

$$\begin{aligned} \int_{\mathbb{R}} f(x)dx &= \int_{\mathbb{R}} \frac{3}{4}x(2 - x)\mathbf{1}_{[0,2]}(x)dx = \int_0^2 \frac{3}{4}x(2 - x)dx = \frac{3}{4} \int_0^2 2x - x^2 dx \\ &= \frac{3}{4} \left( x^2 - \frac{x^3}{3} \right) \Big|_0^2 = \frac{3}{4} \left( 4 - \frac{8}{3} - 0 \right) = \frac{3}{4} \cdot \frac{4}{3} = 1 \end{aligned}$$

both conditions are fulfilled and thus the density is given by

$$f(x) = \frac{3}{4}x(2 - x)\mathbf{1}_{[0,2]}(x).$$

- b) The distribution function is obtained by integrating the density function. For  $x \leq 0$ , this integral equals zero. Therefore, we now assume that  $x \in (0, 2]$ .

$$\begin{aligned} F(x) &= \int_0^x f(y)dy = \int_0^x \frac{3}{4}y(2-y)dy = \frac{3}{4} \int_0^x 2y - y^2 dy = \frac{3}{4} \left( y^2 - \frac{y^3}{3} \right) \Big|_0^x \\ &= \frac{3}{4} \left( x^2 - \frac{x^3}{3} - 0 \right) = x^2 \left( \frac{3}{4} - \frac{x}{4} \right) \end{aligned}$$

According to the results from part a) ii), we have  $F(x) = 1$  for all  $x \geq 2$ . Thus, the distribution function is given by

$$F(x) = x^2 \left( \frac{3}{4} - \frac{x}{4} \right) \mathbf{1}_{[0,2]}(x) + \mathbf{1}_{(2,\infty)}(x),$$

which we were supposed to show.

- c) Now we have to compute the expected value of  $X$

$$\begin{aligned} E[X] &= \int_{\mathbb{R}} x \cdot f(x)dx = \int_{\mathbb{R}} x \cdot \frac{3}{4}x(2-x)\mathbf{1}_{[0,2]}(x)dx = \int_0^2 \frac{3}{4}x^2(2-x)dx = \frac{3}{4} \left( \frac{2}{3}x^3 - \frac{1}{4}x^4 \right) \Big|_0^2 \\ &= \frac{3}{4} \left( \frac{2}{3}2^3 - \frac{1}{4}2^4 \right) = 4 - 3 = 1 \end{aligned}$$

- d) We need to determine the 0.01 quantile of the distribution function  $F$ . If this quantile is greater than 0.15, the manufacturer is not required to sell defective products according to the model. However, if it is less than or equal to 0.15, the manufacturer would be forced to sell inferior goods.

The quantile we are seeking lies within the interval  $[0, 2]$ , where the distribution function is continuous. Thus, we can simply invert the function. We obtain,

$$x^2 \left( \frac{3}{4} - \frac{x}{4} \right) = 0.01 \quad \Leftrightarrow \quad -0.01 + \frac{3}{4}x^2 - \frac{1}{4}x^3 = 0.$$

Hence, we need to find the roots of the polynomial  $-0.01 + \frac{3}{4}x^2 - \frac{1}{4}x^3$  in the interval  $[0, 2]$ . They are quite complex to calculate, so we use R

```
polyroot(c(-0.01, 0, 0.75, -0.25))
```

```
[1] 0.1178063-5.169879e-26i -0.1133486+4.806372e-26i 2.9955423+3.635071e-27i
```

All results contain an imaginary part, but it is of the order  $10^{-26}$ . Therefore, we can consider all three roots as having real (non-complex) values. Since only one root belongs to the interval  $[0, 2]$ , we can conclude that 0.1178063 is the 0.01 quantile. Based on the assumed distribution of  $X$ , the manufacturer appears forced to sell goods that are officially classified as scrap.

*Remark:* We could have also used <http://www.wolframalpha.com/> instead of R.

### Exercise 7.3 (5 points).

Grade all three submissions assigned to you (one being your own) in the peer-review workshop for Problem H.6.1 by June 01, 09:00.

You find the submissions by re-opening the peer-review workshop Problem H.6.1.

**NOTE:** Due to the extended submission deadline for Exercise 6.1 (until May 27, 9:00), you will not be able to start grading before May 27, 12:00.